**Name:** Atharv Revat Shukla

**PRN:25070122053**

**Division:A2**

**Subject:** Programming with Java

**ASSIGNMENT No : 7**

**TITLE :**Implement a Java program to demonstrate inheritance and interfaces using suitable

real-world examples.

**Problem Statement**

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world

examples.

**Learning Objectives**

To implement inheritance and interfaces for achieving code reusability and abstraction.

**Theory**

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and

reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a

contract that implementing classes must follow, enabling abstraction and multiple inheritance of type.

Together, inheritance and interfaces support polymorphism and modular software development. They are

fundamental concepts of object-oriented programming.

**Outcome**

**Conclusion**

This program demonstrates **inheritance** and **method overriding** by calculating the area of different shapes using a common `Shape` class.This program demonstrates

**inheritance** and **interfaces** by using common product properties and displaying details of different product type

**Exercise**

**1.Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.**

```
import java.util.Scanner;

class Shape {
    void area() {
    }
}

class Circle extends Shape {
    double r;

    Circle(double r) {
        this.r = r;
    }

    void area() {
        System.out.println("Area of Circle = " + (3.14 * r * r));
    }
}

class Rectangle extends Shape {
    double l, b;

    Rectangle(double l, double b) {
        this.l = l;
        this.b = b;
```

```java
    }

    void area() {
        System.out.println("Area of Rectangle = " + (l * b));
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter radius: ");
        Circle c = new Circle(sc.nextDouble());

        System.out.print("Enter length: ");
        double l = sc.nextDouble();

        System.out.print("Enter breadth: ");
        double b = sc.nextDouble();

        Rectangle r = new Rectangle(l, b);

        c.area();
        r.area();
    }
}
```

**2. Develop a E-Commerce Product System where electronic, clothing, and grocery products**

**inherit common properties and implement product interfaces.**

```
import java.util.Scanner;

interface Product {
    void display();
}

class Item {
    int id;
    String name;
    double price;

    Item(int id, String name, double price) {
        this.id = id;
        this.name = name;
        this.price = price;
    }
}

class Electronic extends Item implements Product {
    Electronic(int id, String name, double price) {
        super(id, name, price);
    }

    public void display() {
        System.out.println("Electronic: " + id + " " + name + " " + price);
    }
}
```

```
class Clothing extends Item implements Product {
    Clothing(int id, String name, double price) {
        super(id, name, price);
    }

    public void display() {
        System.out.println("Clothing: " + id + " " + name + " " + price);
    }
}

class Grocery extends Item implements Product {
    Grocery(int id, String name, double price) {
        super(id, name, price);
    }

    public void display() {
        System.out.println("Grocery: " + id + " " + name + " " + price);
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Electronic e = new Electronic(1, "Laptop", 50000);
        Clothing c = new Clothing(2, "Shirt", 800);
        Grocery g = new Grocery(3, "Rice", 60);
```

```
        e.display();
        c.display();
        g.display();
    }
}
```